6

Camera Models

A camera is a mapping between the 3D world (object space) and a 2D image. The principal camera of interest in this book is *central projection*. This chapter develops a number of camera *models* which are matrices with particular properties that represent the camera mapping.

It will be seen that all cameras modelling central projection are specializations of the *general projective camera*. The anatomy of this most general camera model is examined using the tools of projective geometry. It will be seen that geometric entities of the camera, such as the projection centre and image plane, can be computed quite simply from its matrix representation. Specializations of the general projective camera inherit its properties, for example their geometry is computed using the same algebraic expressions.

The specialized models fall into two major classes – those that model cameras with a finite centre, and those that model cameras with centre "at infinity". Of the cameras at infinity the *affine camera* is of particular importance because it is the natural generalization of parallel projection.

This chapter is principally concerned with the projection of points. The action of a camera on other geometric entities, such as lines, is deferred until chapter 8.

6.1 Finite cameras

In this section we start with the most specialized and simplest camera model, which is the basic pinhole camera, and then progressively generalize this model through a series of gradations.

The models we develop are principally designed for CCD like sensors, but are also applicable to other cameras, for example X-ray images, scanned photographic negatives, scanned photographs from enlarged negatives, etc.

The basic pinhole model. We consider the central projection of points in space onto a plane. Let the centre of projection be the origin of a Euclidean coordinate system, and consider the plane $\mathrm{Z} = f$, which is called the *image plane* or *focal plane*. Under the pinhole camera model, a point in space with coordinates $\mathbf{X} = (\mathrm{X}, \mathrm{Y}, \mathrm{Z})^\mathsf{T}$ is mapped to the point on the image plane where a line joining the point $\mathbf{X}$ to the centre of projection meets the image plane. This is shown in figure 6.1. By similar triangles, one quickly

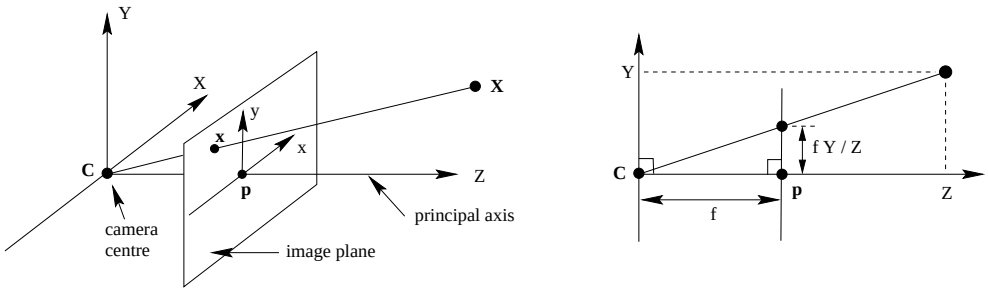


Fig. 6.1. **Pinhole camera geometry.** $\mathbf{C}$ *is the camera centre and* $\mathbf{p}$ *the principal point. The camera centre is here placed at the coordinate origin. Note the image plane is placed in front of the camera centre.*

computes that the point $(\mathrm{X}, \mathrm{Y}, \mathrm{Z})^\mathsf{T}$ is mapped to the point $(f\mathrm{X}/\mathrm{Z}, f\mathrm{Y}/\mathrm{Z}, f)^\mathsf{T}$ on the image plane. Ignoring the final image coordinate, we see that

$$(\mathrm{X}, \mathrm{Y}, \mathrm{Z})^\mathsf{T} \mapsto (f\mathrm{X}/\mathrm{Z}, f\mathrm{Y}/\mathrm{Z})^\mathsf{T} \tag{6.1}$$

describes the central projection mapping from world to image coordinates. This is a mapping from Euclidean 3-space $\mathbb{R}^3$ to Euclidean 2-space $\mathbb{R}^2$.

The centre of projection is called the *camera centre*. It is also known as the *optical centre*. The line from the camera centre perpendicular to the image plane is called the *principal axis* or *principal ray* of the camera, and the point where the principal axis meets the image plane is called the *principal point*. The plane through the camera centre parallel to the image plane is called the *principal plane* of the camera.

Central projection using homogeneous coordinates. If the world and image points are represented by homogeneous vectors, then central projection is very simply expressed as a linear mapping between their homogeneous coordinates. In particular, (6.1) may be written in terms of matrix multiplication as

$$\begin{pmatrix} \mathrm{X} \\ \mathrm{Y} \\ \mathrm{Z} \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} f\mathrm{X} \\ f\mathrm{Y} \\ \mathrm{Z} \end{pmatrix} = \begin{bmatrix} f & & & 0 \\ & f & & 0 \\ & & 1 & 0 \end{bmatrix} \begin{pmatrix} \mathrm{X} \\ \mathrm{Y} \\ \mathrm{Z} \\ 1 \end{pmatrix}. \tag{6.2}$$

The matrix in this expression may be written as $\mathrm{diag}(f, f, 1)[\mathtt{I} \mid \mathbf{0}]$ where $\mathrm{diag}(f, f, 1)$ is a diagonal matrix and $[\mathtt{I} \mid \mathbf{0}]$ represents a matrix divided up into a 3×3 block (the identity matrix) plus a column vector, here the zero vector.

We now introduce the notation $\mathbf{X}$ for the world point represented by the homogeneous 4-vector $(\mathrm{X}, \mathrm{Y}, \mathrm{Z}, 1)^\mathsf{T}$, $\mathbf{x}$ for the image point represented by a homogeneous 3-vector, and $\mathtt{P}$ for the 3×4 homogeneous *camera projection matrix*. Then (6.2) is written compactly as

$$\mathbf{x} = \mathtt{P}\mathbf{X}$$

which defines the camera matrix for the pinhole model of central projection as

$$\mathtt{P} = \mathrm{diag}(f, f, 1)\,[\mathtt{I} \mid \mathbf{0}].$$